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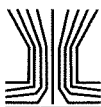
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# Model Calculations and Wind Tunnel Testing of an Isokinetic Shroud for High-Speed Sampling

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**ABSTRACT.** The direction of airflow streamlines and, therefore, particle trajectories around an aircraft will vary during a flight. A coaxial shroud was designed to align airflow and particle trajectories with the tip of a counterflow virtual impactor (CVI), despite deviations in streamlines at the shroud entrance. Both modeling and wind tunnel studies were employed in order to optimize the shroud dimensions for high-speed use on a DC-8 aircraft. A commercial computational fluid dynamics code was used to determine the approximate dimensions necessary to straighten particle trajectories and to maintain isokineticity within the shroud. Characteristics of one shroud then were studied in the U.S. Air Force Academy's Trisonic Wind Tunnel. Flow speeds were measured at various locations within the shroud using small pilot-static tubes. Effects of exit to inlet area ratio and angle of attack on shroud flow speeds were studied, and a final shroud design was chosen based on this information. Model simulations were in general agreement with the wind tunnel results. AEROSOL SCIENCE AND TECHNOLOGY 29:261–280 (1998) © 1998 American Association for Aerosol Research

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## INTRODUCTION

The counterflow virtual impactor (CVI) samples airborne droplets and ice particles from a moving aircraft. These larger cloud particles are separated from interstitial particles (small unactivated particles) and gases in a cloud, and are impacted into a moving airstream inside the CVI tip (Noone et al., 1988). The interstitial particles and gases are rejected by a small airflow (counterflow) out the CVI inlet tip. If impinging streamlines of ambient air are not aligned with the CVI tip,

this may influence the counterflow pattern and allow interstitial particles to be collected. Also, collection efficiency for cloud droplets and ice particles has been observed to decrease when their trajectories are not aligned with the CVI tip (Twohy, 1992), probably as a result of impaction on internal inlet surfaces. The direction of streamlines and particle trajectories around an aircraft will vary throughout the course of a flight due to turbulence and changing flight attitudes—therefore, a single CVI mounting position will not permit ideal collection efficiency.

Shrouds have been used with increasing frequency to align airflow with axes of airborne aerosol and gas-sampling inlets, e.g., McFarland et al. (1989), Soderman et al.

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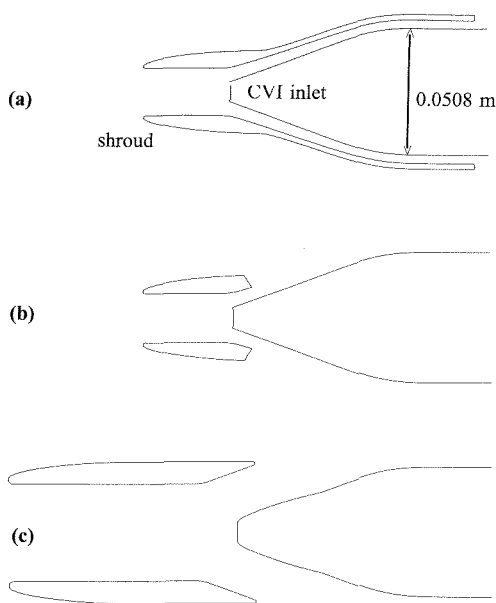
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(1991), Ram et al. (1995), Murphy and Schein (1998), and Eisele et al. (1997). These shrouds have provided significant improvements in flow characteristics and particle collection efficiencies of aerosol inlets. In some cases they also have been used to slow airflow and enhance particle concentrations prior to sampling.

This work describes a shroud that was designed and tested prior to mounting the CVI and shroud on the NASA DC-8 aircraft for contrail and cirrus cloud sampling. The purpose of this shroud is to align airflow and particle trajectories with the CVI inlet tip, despite deviations in streamline direction at the shroud entrance. The needs of the airborne CVI are slightly different from those of the inlets used in the aforementioned studies. Specifically, the following goals were set for this shroud design:

1. Straighten airflow upstream of CVI inlet for incident flow angles up to five degrees. It is important that the airflow itself be aligned with the CVI tip to reduce turbulence and prevent disturbance of the counterflow air (which is designed to prohibit interstitial particles and gases from entering the CVI tip).
2. Maintain approximately isokinetic flow inside the shroud, upstream of the CVI inlet. This will prevent distortion of the particle distribution ahead of the CVI.
3. Substantially straighten trajectories of cloud-sized particles for incident angles up to five degrees under DC-8 flight conditions. This is a challenge, since the DC-8 flies at relatively high airspeeds at high altitudes, and particle stopping distances (and thus their tendency to maintain their initial flow direction) are large. The minimum size of particles that can be sampled by the CVI under these conditions is about  $3\text{ }\mu\text{m}$  aerodynamic diameter, so the shroud must straighten trajectories of particles at least this size to be of use.
4. Keep the shroud small, for mounting practicality and minimal flow distortion downstream. (The CVI was mounted ahead of several other particle sampling



**FIGURE 1.** (a) Initial shroud geometry chosen for simulation. (b) Geometry of modified "cutoff" shroud chosen for simulation based on results for initial geometry. (c) Geometry of final shroud configuration (1.75 AR) chosen for wind tunnel testing based on results of prior simulations. The CVI inlet diameter is 0.0508 m for all geometries.

inlets on the DC-8, and a large shroud could disturb the airflow at their sampling locations). This means that goal (3) cannot necessarily be met for very large particles, since the shroud length required would be prohibitive.

Both modeling and wind tunnel studies were employed in order to optimize the shroud dimensions. A commercial computational fluid dynamics code, STAR-CD (Simulation of Turbulent flow in Arbitrary Regions-Computational Dynamics), was used to test several different configurations (Fig. 1) and determine the approximate dimensions necessary to achieve the above requirements. STAR-CD is supported in the United States by Adapco.<sup>2</sup> The STAR-CD code includes viscous effects and handles compress-

<sup>2</sup>60 Broadhollow Road, Melville, New York 11747, (516) 549-2300.

ible flow and transonic regimes with reasonable accuracy, although prediction accuracy diminishes slightly at or very near actual shocks (Robert Bland, Adapco, personal communication, 1996). Since two-phase flow can be simulated by the STAR-CD code, the straightening effect of the shroud for particle trajectories, as well as for airflow, was examined. (See Laucks and Twohy (1998) for more details of the STAR-CD code and for calculations of particle impaction efficiency within the CVI inlet itself.) Variations in exit area of the final shroud design were tested in a wind tunnel to determine the configuration for which flow through the shroud would be isokinetic and to validate the model.

## COMPUTATIONAL FLOW MODELING

### *Initial Design*

The initial shroud design studied is shown in cross-section in Fig. 1a. The upstream tip of the shroud is shown at the left side of the diagram, while the CVI inlet itself is located downstream of the shroud tip and extends to the right side of the diagram. A modified NACA 1-50-150 airfoil shape was chosen for the shroud exterior, as it does not induce local shocks at DC-8 speeds of Mach numbers ( $M$ ) up to 0.87 (Nichols and Keith, 1951). The initial shroud diameter and length upstream of the inlet were based on dimensions that were previously tested for the Multiangle Aerosol Spectrometer Probe (Baumgardner et al., 1995), and were shown to have good airspeed vs. angle of attack characteristics. The shroud was lengthened slightly (to 0.034 m), however, to accommodate the NACA airfoil shape on the exterior. The inside of the shroud, while slightly rounded at the tip, was basically flat until it reached the plane of the CVI tip. Although less aerodynamic at large incident flow angles, this shape was chosen to minimize compression or expansion of streamlines within the shroud (with possible distortion of particle trajectories).

The amount of flow through the shroud inlet is expected to increase with increasing

cross-sectional area of the shroud exit, up to a limiting value when the flow is choked at a Mach number of 1.0 (Seddon and Goldsmith, 1985). The exact exit area needed for isokinetic flow cannot be simply predicted, however, without knowing the total pressure loss inside the shroud, which may be complicated by the presence of a shock in this transonic flow regime. Note that the CVI inlet itself presents a substantial blockage to the airflow and that the ratio of boundary layer area to total area is expected to be large in the narrow channel formed between the shroud and the CVI inlet. Therefore, the shroud exit area (the area of the annulus formed between the outside of the CVI inlet and the interior of the shroud at its back) was set initially at twice its inlet area to prevent slowing in the region upstream of the CVI inlet.

The flow field was assumed to be axisymmetric, so the computational mesh was constructed for only half ( $180^\circ$  from top to bottom) of the shroud and inlet. Three-dimensional cells were used to describe the domain comprising the airflow around and through the shroud. The number of cells varied from 56,000 to 148,000, depending on the dimensions of the shroud being studied, since enhanced resolution was used near the shroud itself and in regions where fine flow details were expected to affect the overall airflow through the shroud. The plane formed by a vertical slice through the longitudinal axis of the model was defined as a symmetry plane (a surface along which the normal velocity component is zero). A cross-section of a portion of this plane, showing the mesh near the shroud, is shown in Fig. 2.

The upstream model boundary, 0.10 m ahead of the shroud tip, initially was set with the following conditions: an air density of  $0.50 \text{ kg m}^{-3}$  (314 mb, 217 K) and Mach number of 0.65, at the low end of research speeds flown in the upper troposphere by the DC-8. The downstream boundary, 0.20 m behind the shroud tip, was defined in terms of static pressure relative to the upstream boundary. After calculating the flow field using several different values, it was determined that a pressure drop of 5 mb was appropriate in

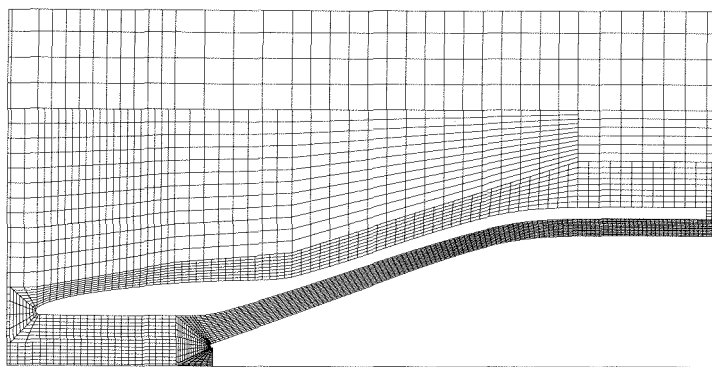


FIGURE 2. Cross section of cells comprising a portion of the model domain for the initial shroud configuration (Fig. 1a).

order to maintain the proper upstream boundary conditions. For all the models in this study, STAR-CD used linearization of nearest neighbor values to approximate the convective and diffusive terms of the equations of motion and the Semi-Implicit Method for Pressure Linked Equations (SIMPLE) (Patankar and Spalding, 1972) to achieve convergence. A  $k$ - $\epsilon$  turbulence model (Launder and Spalding, 1974) was employed, with turbulence parameters at the upstream boundary set at 1% intensity and 1 mm length scale (as recommended by Adapco). The internal CVI flow was not modeled, but a small counterflow rate out the CVI inlet tip was simulated at the CVI tip boundary.

A contour plot of Mach numbers predicted for the first shroud design is given in Fig. 3a. Note the slowing of the flow ahead of the shroud and the CVI inlet tip, and the rapid acceleration of the flow around the CVI tip. A strong shock wave is actually predicted within the shroud slightly behind the CVI tip. The acceleration and the likelihood of exceeding sonic velocities for high Mach flow forced around an obstacle (the CVI tip) was predicted by the Nichols and Keith work on engine intakes (where the obstacle was the spinner within the engine housing). Here, the high speeds are apparently exacerbated by the relatively sharp angle inside the shroud near the CVI tip and the narrow channel through which the air

must pass. Behind the shock wave, the static pressure was predicted to increase, the total pressure to decrease, and the flow to slow dramatically. This is expected from compressible flow theory (e.g., Shevell, 1989). In response to this adverse pressure gradient, the boundary layer apparently becomes turbulent behind the shock. This is evidenced by a substantial increase in thickness and turbulent kinetic energy (not shown) of the boundary layer.

Although this initial shroud geometry did not appear to be optimal, the flow speed within most of the shroud ahead of the CVI tip was predicted to be within 2% of isokinetic. Therefore, this first simulation was used to predict the impact of the shroud on off-axis streamlines and particle trajectories launched near the upstream model boundary. The same model parameters were used, except the flow was initialized with a five-degree angle at the upstream boundary. The lower solid line in Fig. 4 shows the trajectory of a unit-density, 1  $\mu\text{m}$  diameter particle superimposed on the predicted flow vectors. Note the slight angle in the vectors upstream of the shroud and the asymmetric flow pattern at the shroud inlet and within the shroud boundary layer. The shroud apparently does align both the airflow and the trajectories of this relatively small particle with the longitudinal axis of the CVI inlet. However, when a 10  $\mu\text{m}$  particle is considered (upper trajectory in Fig. 4), the shroud

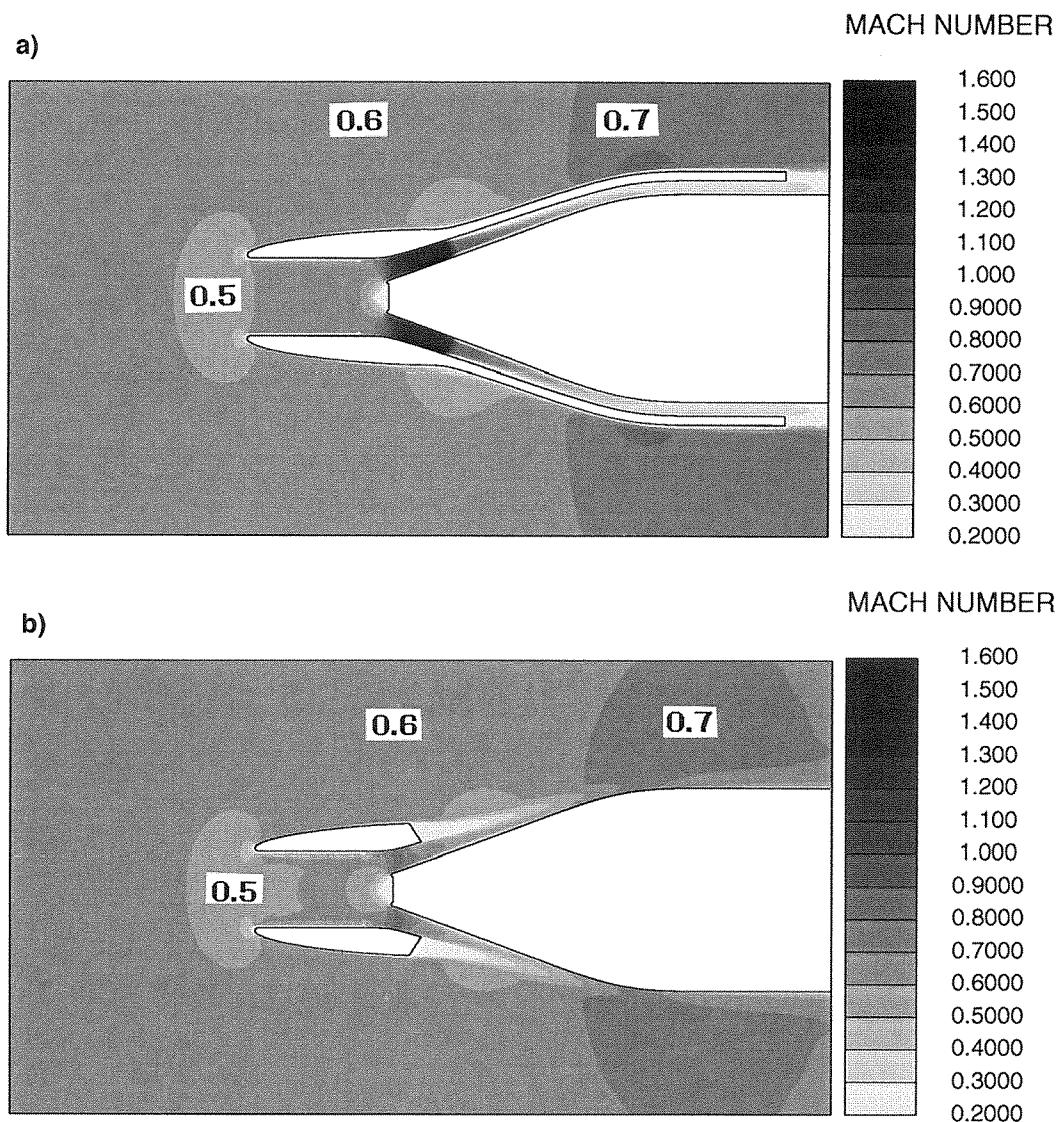


FIGURE 3. (a) Contour plot of Mach number predicted by the model for geometry of Fig. 1a. (b) Contour plot of Mach number predicted by the model for cutoff shroud (Fig. 1b). The Mach number at the upstream boundary (not shown) is 0.65 for both cases. The numeric labels indicate the bottom of the contour value range.

is of insufficient length to straighten the particle's path before it reaches the CVI inlet. The relatively large airspeed of this high-altitude application means particles have more resistance to changing direction than they would in a low altitude, low airspeed application.

### Cutoff Shroud

This model had the same basic geometry as the previous one, except that the back end of the shroud was cut off so that the airflow around the CVI inlet would not be confined to such a narrow channel (Fig. 1b). This also

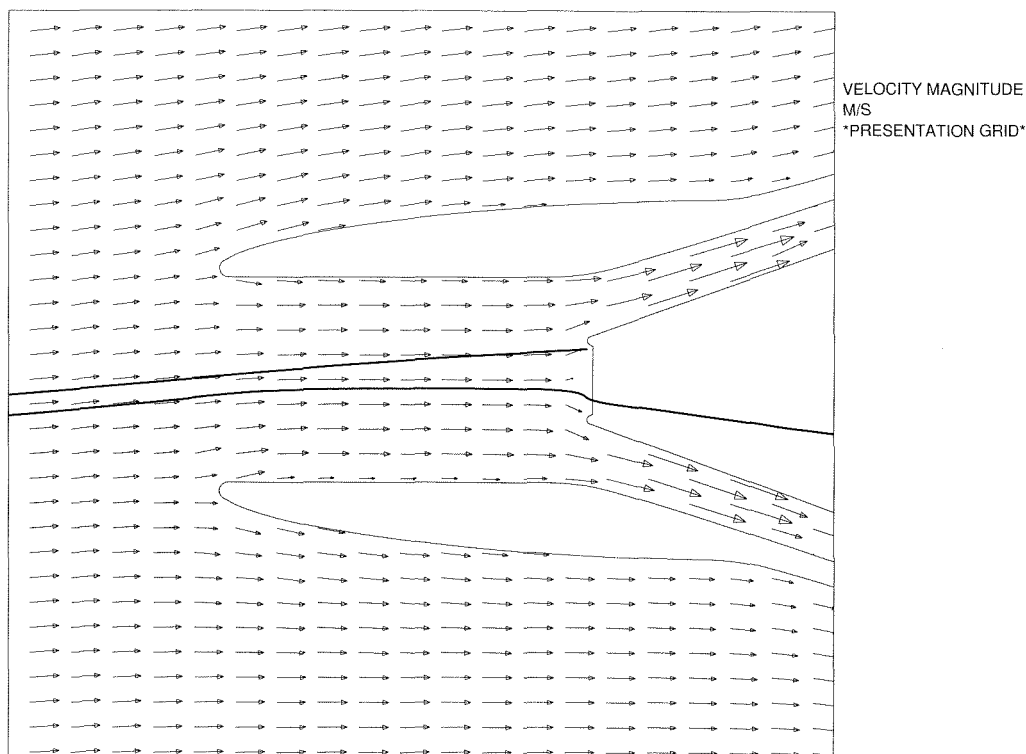


FIGURE 4. Trajectories of  $1\ \mu\text{m}$  and  $10\ \mu\text{m}$  particles (lower and upper trajectories, respectively) superimposed on velocity vectors predicted for a  $5^\circ$  angle of attack for geometry of Fig. 1a. Vector lengths are proportional to velocity magnitude, which is  $194\ \text{m s}^{-1}$  (Mach = 0.66) at the center of the shroud upstream of the CVI inlet tip.

reduced the ratio of shroud exit to inlet area from 2.0 to about 1.1. In the interests of time, no attempt was made to adjust the upstream dimensions of the shroud or to smooth the back end to make it aerodynamic; this was simply a test to determine if the cutoff shroud configuration would reduce the severity of the shock wave. This appears to work, as Fig. 3b attests. The flow still accelerates just downstream of the CVI inlet, but it slows again near the back of the shroud and never exceeds Mach 1. In addition, the flow within the shroud is now subisokinetic (relative to the upstream Mach number) by about 11%. Just behind the blunt back of the shroud, the streamlines (not shown) cannot follow the shroud contours, and the flow was observed to separate in the downstream wake. The abrupt,

nonaerodynamic angles at the back of the shroud were rounded slightly in the final design to alleviate this problem.

#### *Final Shroud Design: 1.75 Area Ratio*

A modified shroud design that incorporated information obtained from both of the previous simulations was constructed subsequently for testing at the U.S. Air Force Academy's Trisonic Wind Tunnel. The cutoff shroud design was employed, but with some variations, shown in Fig. 1c. The straight front end of the shroud was lengthened to 0.076 m to improve its trajectory-straightening capabilities; it was also widened to 0.038 m inside diameter in order to

maintain about the same length/width ratio and to reduce the blocking effect of the CVI inlet. The back end of the shroud was rounded to make it more aerodynamic. Finally, the exterior of the CVI inlet (inside the shroud) was rounded slightly to allow the airflow to smoothly follow the CVI exterior surface. The CVI tip itself was kept relatively sharp to minimize its outside radius and thereby allow smaller particles to impact on the CVI tip.

Several variations of shroud exit to inlet area ratios (AR) were tested in the wind tunnel by moving the shroud forward or back relative to the CVI inlet itself. The inlet area is simply the cross-sectional area of the circle formed by the shroud leading edge, while the exit area is the annular area formed between the shroud interior and the CVI inlet exterior at the back of the shroud. The shroud and inlet configuration with the largest AR (1.75) was modeled using the STAR-CD code. The wind tunnel conditions (total pressure of 1515 mb, total temperature of 300 K, and Mach 0.805) corresponded to a velocity of  $265 \text{ m s}^{-1}$ , static pressure of 991 mb, and static temperature of 265 K. These values were used for upstream boundary conditions in the simulation. For consistency with the wind tunnel testing, no counterflow was included in this model.

The predicted Mach number field is given in Fig. 5a. For this large exit area, the model predicts not only that the flow is super-isokinetic, but that a shock occurs within the main portion of the shroud, upstream of the CVI inlet. Just behind the shock, the Mach number abruptly drops, and pressure and density increase. The shock is induced by the angle near the rear of the shroud, where the flow must abruptly change direction and become subsonic again before it reaches the trailing edge of the shroud. The situation is analogous to transonic flow over an airfoil. Further increases in exit area would not be expected to increase the Mach number in the shroud, since the flow is already effectively choked.

### ***Final Shroud Design: Area Ratio 1.34***

The final shroud configuration was determined by the wind tunnel measurements (discussed in the following section) to be approximately isokinetic for an AR of 1.34. This AR was modeled first for the wind tunnel conditions, and then for the actual high-altitude flight conditions of the DC-8 (Mach 0.805, 200 mb, 217 K, and  $0.3 \text{ lit min}^{-1}$  volumetric counterflow rate). This allowed particle trajectories to be calculated for the actual flight conditions.

The predicted Mach number field for the AR 1.34 shroud was similar under both wind tunnel and DC-8 conditions, indicating that Mach similarity was successfully achieved. Figure 5b shows the predicted Mach number under DC-8 conditions. A shock is still predicted where the air accelerates around the angle near the rear of the shroud, but the shock is less intense and does not extend upstream of the CVI inlet. The predicted Mach numbers within the shroud are significantly less than those predicted for the larger AR (Fig. 5a), and only slightly larger (about 4%) than the isokinetic value of 0.805.

This same model was then initiated with a  $5^\circ$  incident flow angle, in order to examine the flow-straightening properties of the shroud. As observed with the initial shroud design, the airflow straightens shortly after it enters the shroud (Fig. 6a). When the same model was run with a  $10^\circ$  angle (not shown), the shroud still appeared to straighten the airflow and flow separation did not occur. At a  $16^\circ$  angle, however, flow separation and a recirculation zone was predicted downstream of the shroud tip (Fig. 6b).

Several sizes of unit-density, spherical particles (representing droplets or ice crystals with these aerodynamic diameters) were launched at the upstream boundary and followed through the model domain for the  $5^\circ$  and a  $2.5^\circ$  case. The conditions tested here were near the maximum altitude and air-speed of the DC-8, and the predicted speeds within the shroud were slightly higher than those implied by the wind tunnel testing. These results are therefore conservative in terms of the ability of the shroud to



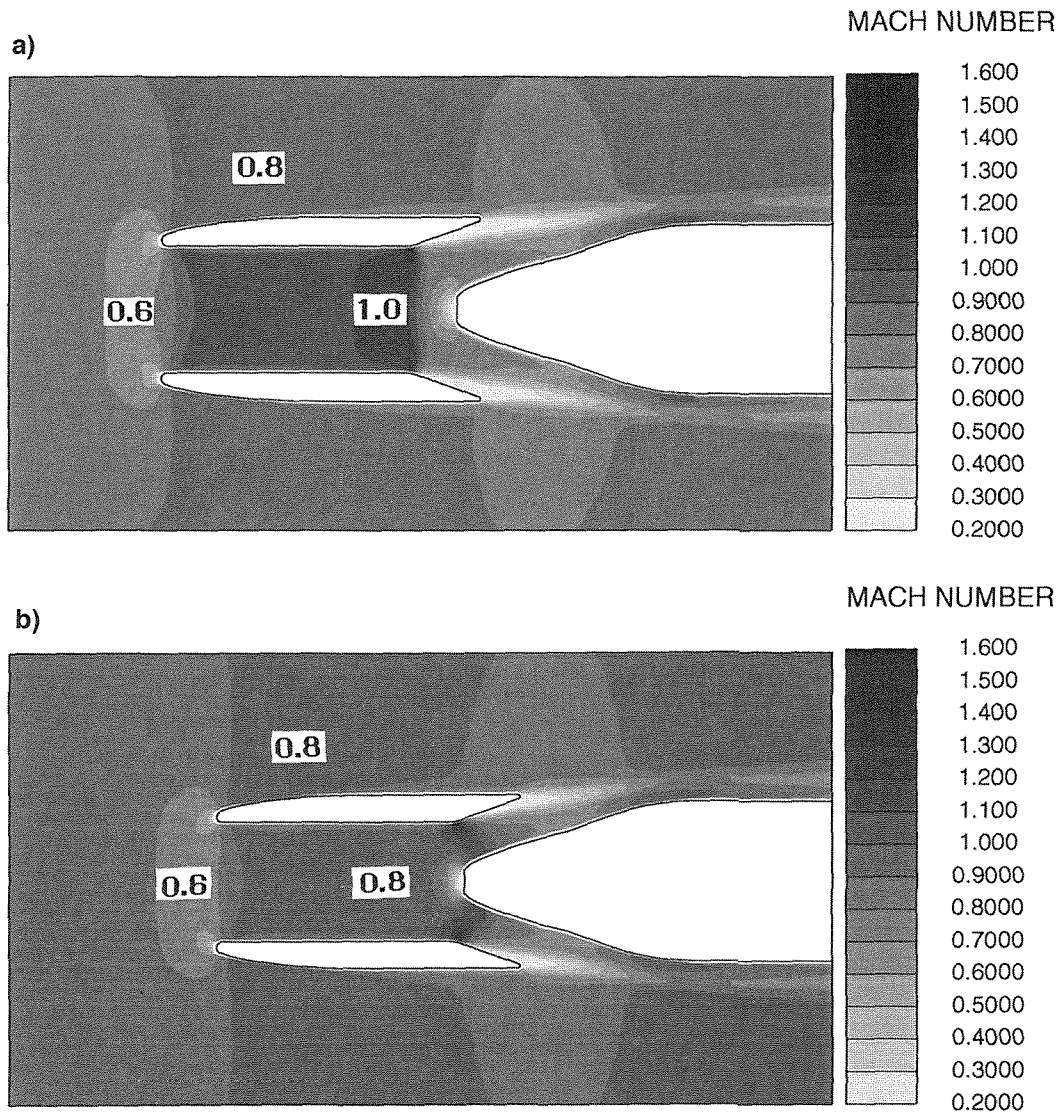
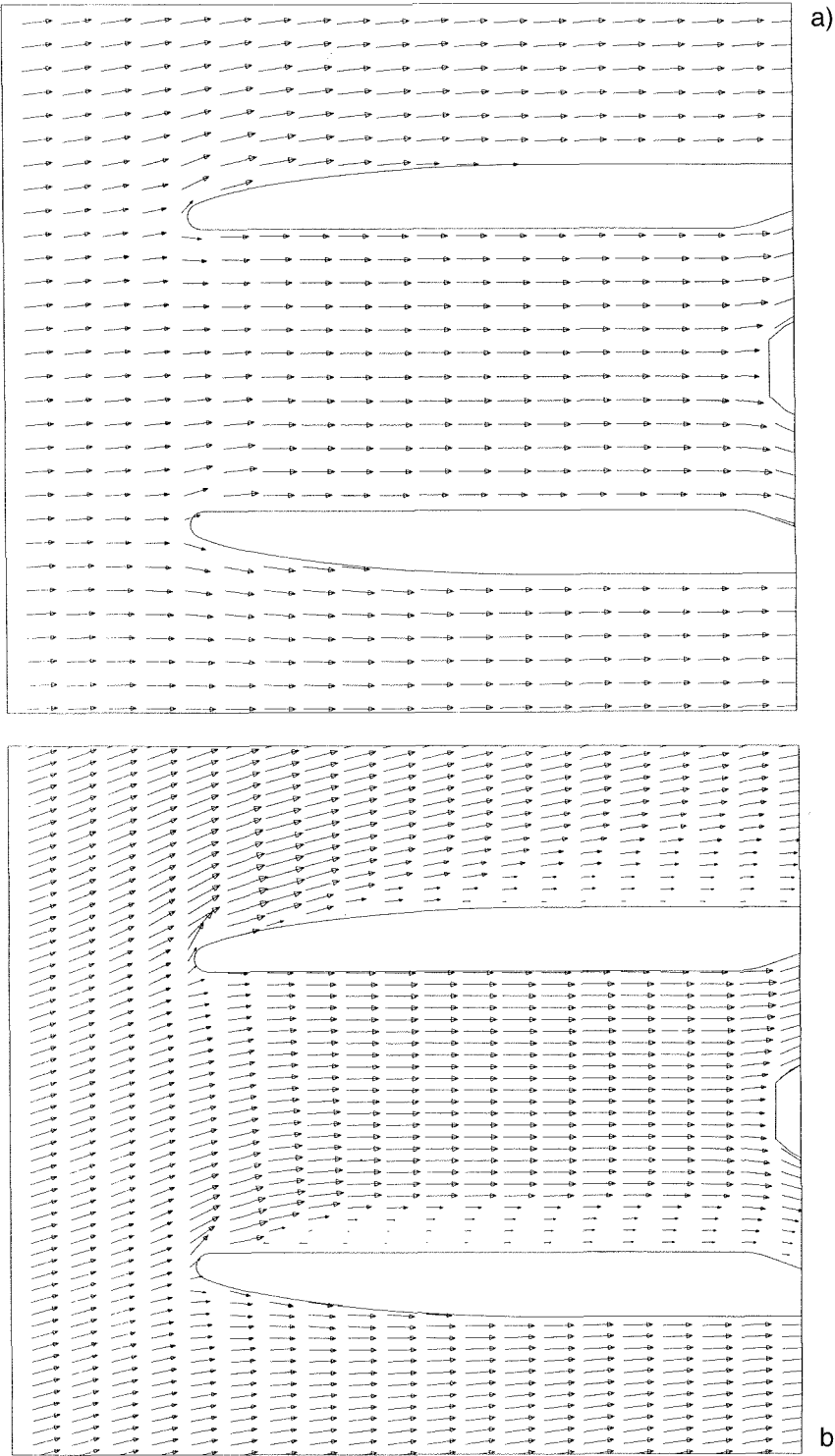


FIGURE 5. (a) Mach number predicted by the model for geometry of Fig. 1c with an area ratio of 1.75, for wind tunnel conditions (Mach 0.805). (b) Mach number predicted by the model for shroud shown in Fig. 1c, but with an area ratio of 1.34, for DC-8 high-altitude operating conditions (Mach 0.805). The numeric labels indicate the bottom of the contour value range.

straighten particle trajectories, since at lower altitudes and airspeeds, particles would be more likely to follow streamlines. Trajectories of various sized particles (1.0,

2.5, 5.0, 7.5, 10.0, 15.0, and 20.0  $\mu\text{m}$  diameter) are shown in Fig. 7 for the  $5^\circ$  incident angle. The shroud appears to straighten 1.0, 2.5, 5.0, and 7.5  $\mu\text{m}$  sized particles, the four



**FIGURE 6.** (a) Flow straightening effect of shroud at a 5° incident angle under DC-8 operating conditions. Velocity magnitude is 250 m s<sup>-1</sup> (Mach = 0.85) at the center of the shroud upstream of the CVI inlet tip. (b) Flow separation predicted at a 16° incident angle. Velocity magnitude is 260 m s<sup>-1</sup> (Mach = 0.89) at the center of the shroud upstream of the CVI inlet tip. Vectors are plotted more densely in (b) to increase resolution in separation zone.

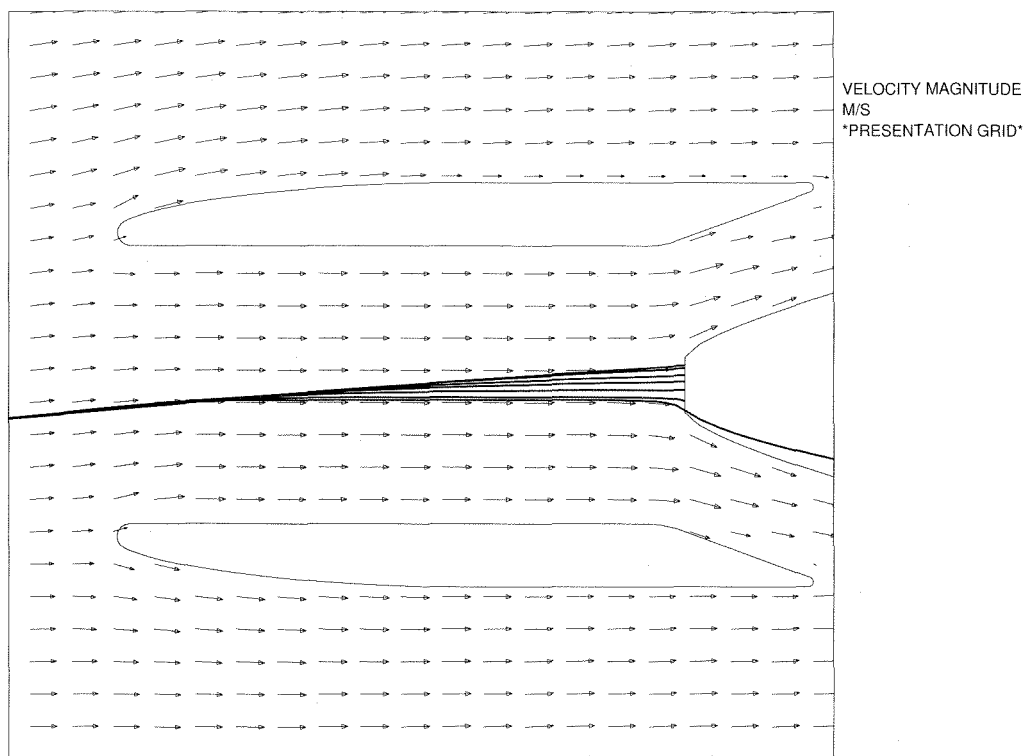


FIGURE 7. Predicted trajectories of 1.0, 2.5, 5.0, 7.5, 10.0, 15.0, and 20.0  $\mu\text{m}$  particles that were begun at the same point for the final shroud configuration (DC-8 conditions) at a  $5^\circ$  incident angle. The 1.0  $\mu\text{m}$  particle terminates as the lower-most trajectory, while the trajectory of the 20.0  $\mu\text{m}$  particle is uppermost. Velocity magnitude is  $250 \text{ m s}^{-1}$  (Mach = 0.85) at the center of the shroud upstream of the CVI inlet tip.

lowest trajectories shown, for the  $5^\circ$  case (and up to  $10 \mu\text{m}$  in size for the  $2.5^\circ$  case, not shown). The angles computed from the trajectories reveal, however, that there is still a small residual angle (less than  $1^\circ$ ) for the  $7.5 \mu\text{m}$  particle as it approaches the CVI tip. Also, even the smallest particles are not straightened completely (to  $0.0 \text{ m s}^{-1}$  vertical velocity) until they have traveled about three-quarters of the distance down the shroud. Thus the straightening is really an exponential process: the majority of it occurs in a relatively short distance, but a much longer distance is required to completely straighten the trajectory. The exponential nature of this process is shown in Fig. 8, where the distance required to straighten the trajectory to various residual angles is plot-

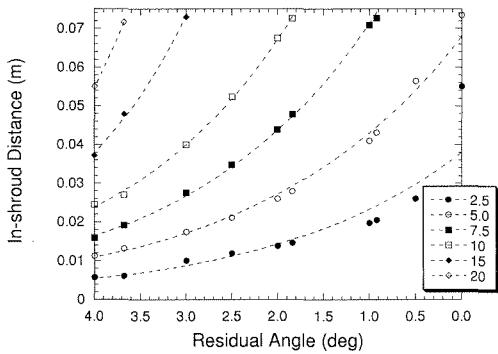
ted. This behavior is explored further and generalized in a later section.

The trajectories in Fig. 7 indicate that while  $1.0 \mu\text{m}$  particles follow the streamlines around the CVI tip,  $2.5 \mu\text{m}$  and larger particles are expected to impact on the CVI tip. Using cut size parameterizations developed by Laucks and Twohy (1998) for another CVI, droplets larger than about 2 or  $3 \mu\text{m}$  diameter are expected to be collected for these conditions, in agreement with the model predictions.

## WIND TUNNEL TESTING

### Experimental Design

The final shroud design was tested in the Trisonic Wind Tunnel (designed for sub-



**FIGURE 8.** Predicted distance required to straighten particles of various sizes to given residual angles, with a 5° incident angle (Mach = 0.805, 200 mb). (The deviation from exponential behavior for a 2.5  $\mu\text{m}$  droplet with a residual angle of 0.0 is due to its tendency to follow airflow streamlines. While the streamlines closely follow shroud contours and become nearly straight in a relatively short distance, they require almost the full length of the shroud to achieve perfect alignment with the shroud axis. This may be due to the slightly thicker boundary layer on the lower shroud surface.)

sonic, transonic, and supersonic flows) at the U.S. Air Force Academy in Colorado Springs, Colorado. A shroud and mock CVI inlet with a solid tip were fabricated out of aluminum. The shroud was attached with four thin horizontal struts to the exterior of the CVI inlet. The CVI mock-up was centered in the 0.30 m  $\times$  0.30 m tunnel cross section, and blocked less than 2% of the tunnel through area. The CVI replica, including the shroud, extended 0.36 m into the 0.91 m tunnel test section. The tunnel was run at a Mach number of 0.805, about that expected at the DC-8's highest altitude.

The length of the struts attaching the shroud to the main CVI inlet was varied, allowing the shroud to be moved either away from the inlet (creating a large annular exit area) or toward the inlet (small exit area). In this way, the effects of different exit to inlet area ratios could be studied using only one shroud. Three shroud area ratios (1.75, 1.50, and 1.25) were actually tested.

For each AR configuration, two pitot-static tubes were used to measure the flow speed along the shroud centerline and near its lower edge, at different distances back

**TABLE 1.** Mach Numbers Measured within CVI Shroud in Wind Tunnel

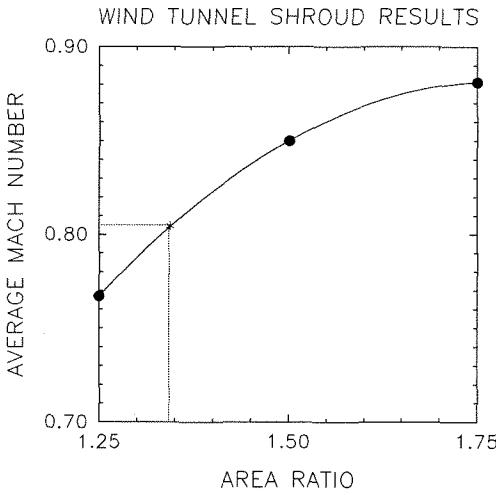
| RUN # | Upstream     |               |             |
|-------|--------------|---------------|-------------|
|       | Distance (m) | Mach (center) | Mach (edge) |
| 2A    | 0.0595       | 0.756         | 0.770       |
| 2B    | 0.0437       | 0.766         | 0.775       |
| 2C    | 0.0159       | 0.779         | 0.794       |
| 3A    | 0.0635       | 0.828         | 0.834       |
| 3B    | 0.0476       | 0.844         | 0.858       |
| 3C    | 0.0159       | 0.879         | 0.888       |
| 4A    | 0.0675       | 0.856         | 0.861       |
| 4B    | 0.0516       | 0.880         | 0.883       |
| 4C    | 0.0159       | 0.907         | 0.924       |

within the shroud. Locations (A, B, C) where the static pressure measurements were taken during these runs are shown in Fig. 9. (Since the pitot-static tubes were moved manually between runs, a limited number of positions were chosen due to time constraints in the tunnel.) The diameter of the pitot-static tubes was only 0.003 m, so each one blocked less than 1% of the shroud cross-sectional area. For each of the pitot-static tube positions, the entire shroud assembly was inclined upward by 3.3°, 6.7°, and 10.0° (relative to the tunnel axis) in order to simulate progressively increasing angles of attack. A set of measurements also was taken without the shroud with the pitot-static tubes well upstream of the CVI inlet to verify that our pressure sensors agreed with those used by the Air Force to determine the reference tunnel Mach number.

**Mach Number vs. Position and Area Ratio**

Table 1 summarizes the Mach numbers calculated from the pitot-static measurements for Runs 2 (small AR) through 4 (large AR). Positions in the table (upstream distance) are referenced relative to the CVI inlet tip. Positions A and B were fixed relative to the shroud itself for the experiment and thus varied relative to the inlet tip depending on area ratio, while position C was 0.0159 m ahead of the inlet tip for all ARs. For all three configurations, Mach number in-





**FIGURE 10.** Average Mach numbers measured within the shroud for the three area ratios chosen. Final configuration for isokinetic flow (chosen for the DC-8 aircraft probe) is marked with an \*.

creased with distance downstream in the shroud, both along the centerline and nearer the edge. This increase was between 3–6%, depending on the run. The edge flow was slightly faster than the center flow for all positions.

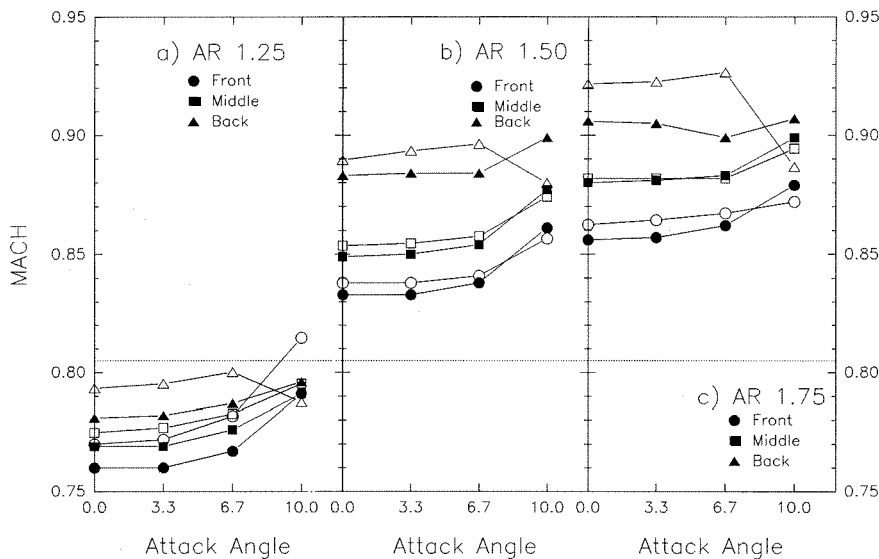
One purpose of the wind tunnel tests was to determine the exit area for which the flow speed within the shroud was approximately isokinetic. Since the flow speed varied within the shroud, the Mach number at the three positions A, B, and C were averaged for each of the exit areas and compared with the tunnel Mach number (0.805). The resulting averaged Mach numbers for the three positions are shown in Fig. 10. It is apparent that the flow is subisokinetic within the shroud for the smallest AR (1.25) and superisokinetic for the two larger ARs. The flattening of the Mach number vs. area ratio curve at high AR is expected, since the amount of flow through the shroud will asymptote to a limiting value as the flow becomes choked at large exit areas (Seddon and Goldsmith, 1985, Fig. 1.3). Using Fig. 10, a second-order regression was implemented to select an AR corresponding to isokinetic flow. Since the target Mach number was 0.805, an AR of

1.34 was used for the shroud ultimately flown on the DC-8.

### *Effect of Angle of Attack*

Dependence of Mach number on angle of attack was tested for all exit areas and all the pitot-static tube positions. Based on aerodynamic theory as well as our modeling work, certain predictions can be made regarding expected flow speed changes with angle of attack. Specifically, we expect Mach number to be approximately constant (or even to increase slightly) with increasing angle of attack for airflow that smoothly follows the shroud interior shape. A slight speed increase may be observed since the airflow at certain locations must actually accelerate to circumvent the shroud when the airflow is impinging at an angle (Fig. 6a). In contrast, if the incident angle is so large that the flow cannot smoothly follow the shroud contours, the flow may separate, lose energy, and slow downstream of where the separation occurs. Figure 6b shows this situation at the lower shroud edge for a 16° incident angle, where Murphy and Schein (1998) observed flow separation in a similar shroud. Thus Mach number will stay relatively constant or increase slightly with increasing angle of attack for smooth, well-behaved flow, and will decrease with increasing angle of attack once flow separation occurs.

The measured mean Mach number for different exit areas and pitot locations is plotted in Fig. 11. Results for small to large area ratios (Figs. 11a–c) are plotted on the same vertical axis, and the differences in Mach number at a 0.0° angle demonstrate the effects of area ratio discussed earlier. Mach number remains about the same or increases slightly with increasing angle of attack for all positions except the back measurement (position C) along the shroud edge (hollow triangles), where Mach number drops sharply at a 10° angle for all exit areas. Also, for the same position, fluctuations in the magnitude of differential pressure (measured at 3 Hz, not shown) more than doubled at 10° relative to the other angles. This indicates that while the flow within the



**FIGURE 11.** Dependence of Mach number on angle of attack, as measured in the wind tunnel. Solid symbols are center measurements, hollow symbols are measurements at lower shroud edge. “Front,” “middle,” and “back” refer to locations A, B, and C, respectively, along the shroud axis. Dotted line represents isokinetic conditions (relative to freestream Mach number of 0.805). (a) 1.25 area ratio. (b) 1.50 area ratio. (c) 1.75 area ratio.

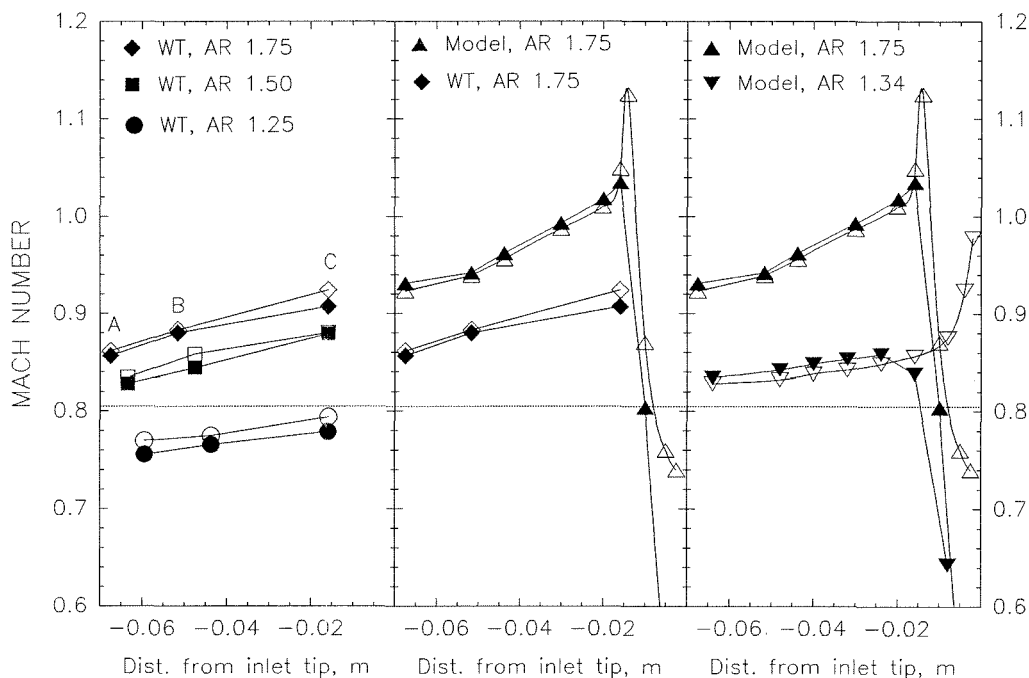
shroud is well-behaved at angles of attack up to 6.7°, it separates and becomes turbulent along the shroud edge at angles between 6.7° and 10.0° for all the area ratios tested. The resulting flow disturbance apparently occurs downstream of the front and middle measurement positions but does not propagate from the shroud edge inward to the shroud centerline, from which droplets are sampled.

These results indicate that flow separation occurs in this shroud at smaller angles than the 17° quoted for the shroud used by Eisele et al. (1997) and the 16° for Murphy and Schein (1998) shroud. The less-aerodynamic inner surface of this shroud (used to prevent distortion of particle trajectories) and the higher Mach numbers within the shroud probably account for this discrepancy. Since deviations in airflow angles into the probe during normal flight conditions are not expected to be more than a few degrees, the flow-straightening capabilities of this shroud are adequate for our application. This is in contrast to Eisele et al. (1997) shrouded gas inlet, which was intentionally mounted at a

10° angle to the mean flow in order to exclude cloud droplets from the inlet.

### COMPARISON OF COMPUTATIONS AND MEASUREMENTS

A summary of both computations and wind tunnel measurements for the final shroud configuration is presented in Fig. 12. The dotted line at 0.805 Mach represents isokinetic flow. The first panel shows the wind tunnel results for the three ARs tested, the second panel compares the model to the wind tunnel measurements for 1.75 AR, and the third panel compares the two model runs at 1.75 and 1.34 AR. The second panel shows that the 1.75 AR computational results do predict the qualitative nature of the flow within the shroud as measured in the wind tunnel tests: namely, that there is little variation in Mach number radially across the shroud (outside the boundary layer) and that the flow speed increases with distance downstream throughout the shroud. However, the model overestimates the absolute value of



**FIGURE 12.** Overall comparison of Mach numbers, both measured and modeled for various shroud area ratios. The abscissa is distance ahead of the CVI tip, in the first panel, corresponding to points A, B, and C in the wind tunnel. Solid symbols represent values along the shroud centerline, while hollow symbols are near the edge. The first panel shows the measured results of the 1.25, 1.50 and 1.75 AR wind tunnel (WT) run. The second panel compares the model with the wind tunnel results for 1.75 AR, and the final panel summarizes the results of the 1.75 and 1.34 AR model runs.

Mach number relative to the wind tunnel results by about 8% upstream of the shock, and by 14% in the predicted shock region. In addition, the flow speeds at the edge pitot locations are predicted to be slightly slower than the center flow speeds, while the measurements indicate that the edge flow was slightly faster than flow along the centerline. However, these differences are within the uncertainty of the measurements (discussed later).

For the 1.34 AR shroud (third panel), the model predicts speeds only about 4% higher than isokinetic for positions similar to those tested in the wind tunnel. Thus, the model appears to reflect reality more accurately when the shock is less pronounced (see Fig. 5).

The wind tunnel measurements indicated flow separation occurred at angles of attack

between 6.7° and 10.0°. However, while separation was observed with a 16° simulation (Fig. 6b) the model did not predict flow separation at a 10° angle. Possible reasons for these differences between model and measurements are discussed in the next section.

Several sensitivity tests were run to see if better agreement between computational results and measurements could be obtained by changing model parameters. These included varying the turbulence model (standard  $k-\epsilon$  vs. renormalization group  $k-\epsilon$ ), varying turbulence intensity and length scale, and varying downstream boundary conditions. The simulation also was run with and without the attachment struts, and with and without the central pitot tube. Differences in the predicted Mach number for all these simulations were less than 1%. The



Gamma solution method, a filtered central differencing scheme, has been shown to produce the most realistic results for transonic flow over an airfoil (R. Bland, Adapco, personal communication, 1996). However, the filtering also increases the difficulty of convergence, and simulations using this solution method failed to converge.

## UNCERTAINTY ANALYSIS

### Wind Tunnel Measurements

Mach number ( $M$ ) is calculated from the ratio of the total pressure,  $p_T$ , to static pressure,  $p_S$ , (Shevell, 1989, Eq. 7.21):

$$M = \sqrt{\frac{2}{\gamma - 1} \left[ \left( \frac{p_T}{p_S} \right)^{(\gamma-1)/\gamma} - 1 \right]},$$

where  $\gamma$  is the ratio of specific heats,  $c_p/c_v$ . Thus the primary sources of uncertainty in Mach number determined from the wind tunnel measurements are the Motorola MPX transducers used to determine the static and total pressures. These have an overall (bias and random) uncertainty of 1.8%. The propagation of errors method (Bevington and Robinson, 1992) was used to estimate the total uncertainty in Mach number derived from the wind tunnel measurements, namely

$$\sigma_M^2 \equiv \left( \frac{\partial M}{\partial p_T} \sigma_{p_T} \right)^2 + \left( \frac{\partial M}{\partial p_S} \sigma_{p_S} \right)^2,$$

where  $\sigma$  is the uncertainty in the subscripted parameter. Using this technique and using typical  $p_T$  and  $p_S$  values of 1515 mb and 915 mb, respectively, we obtain an overall uncertainty for the wind tunnel Mach number of 0.024, 3.0% of the freestream value.

### Computational Flow Analysis

The accuracy of the computational results depends on a variety of factors, including the differencing scheme used, the turbulence model and parameters, and the resolution of the computational mesh. Relative to the wind tunnel measurements at the large AR, the model exhibited speeds that were higher

(by 8–14%) within the shroud, slightly slower speeds near the shroud edge, and a lack of flow separation at  $10^\circ$ . Given that the Mach numbers measured in the wind tunnel have an uncertainty of only 3%, this implies additional uncertainties in the model.

These observed differences between model and measurements suggest that the boundary layer region within the shroud is not predicted correctly by the model. In fact, although 30 cells were used across the diameter of the shroud, the “ $y^+$ ” values (dimensionless normal distance from the wall) for cells along the shroud internal wall were about 300, substantially higher than the 30–200 values recommended for optimal prediction of a turbulent boundary layer. This essentially means that the mesh is coarser than optimal in the near-wall boundary layer region, and the accuracy of the boundary layer prediction may be diminished. This is the most likely cause of the difference between the model and measurements, although difficulty in precisely modeling the shock region may contribute as well. For future models, a finer mesh in the boundary layer and in potential shock regions should be constructed.

## EXTRAPOLATION OF RESULTS

This shroud was designed for the rather specific transonic flow regime of the DC-8 and for the unusual flow configuration of the CVI. Still, certain generalizations can be made that may aid in the design of other shrouds for airborne measurements.

### Flow-Straightening Properties

It is expected that the flow-straightening capabilities of this shroud will apply for lower speed conditions as well. The flow may, in fact, be less likely to separate at lower freestream speeds, as work by others (e.g., Murphy and Schein, 1998) has indicated. Although the tip of this shroud was rounded slightly, an airfoil shape was not employed for the shroud interior, since this might have distorted airflow streamlines and particle trajectories. Since flow separation did not

occur until incident angles of  $6.7^\circ$  to  $10.0^\circ$  were encountered, an airfoil-shaped shroud interior apparently is not required unless larger incident angles are encountered.

### *Achievement of Isokinetic Flow*

The exit to inlet area ratio determined here for isokinetic flow will not apply necessarily to other airborne shrouds, for the following reasons. First, the presence of a shock means that the flow has been "choked," or has achieved its maximum flow rate at a Mach number of 1.0, in the region upstream of the shock. At much lower initial speeds, no shock will occur, and using the 1.34 area ratio determined here, the speed through the shroud may become super-isokinetic. In fact, when the model was run at a lower speeds representing medium and low altitude operations (Mach 0.54 and 0.29, respectively), the mean flow within the shroud was predicted to be about 22% super-isokinetic (vs. 4% super-isokinetic predictions at Mach 0.805). The second reason the 1.34 AR may not apply to other shrouds is that the relatively stagnant air at the CVI tip presents a greater blockage to the flow than a more standard inlet would. The effect of this blockage is not known precisely, but probably means that a larger AR is required to achieve isokinetic flow for the CVI shroud than for shrouds around flow-through inlets. An area ratio slightly larger than 1.0 is expected to be required in all cases, to allow for pressure losses in nonaerodynamic features such as the shroud boundary layer and the exit geometry. Thus for a subsonic isokinetic shroud (with a flow-through inlet downstream), we would expect that an AR slightly larger than 1.0, but less than 1.34, would be required.

### *Particle Trajectories*

The final way that these results may be extrapolated for the benefit of other researchers relates to particle behavior in the shroud. Particle trajectories presented in the previous section are specific to the angles we investigated and to the particular environ-

mental conditions of the high-altitude DC-8. However, a more general prediction of length required to straighten particles of various sizes in an isokinetic shroud under different environmental conditions can be made using dimensional analysis and some knowledge of aerosol physics, as follows.

The distance required to straighten particle trajectories,  $L$ , is expected to be a function of the initial trajectory angle  $\varphi$ , the particle and airstream velocity, and particle relaxation time,  $\tau$ , where

$$\tau = d_p^2 \rho_p / (18 \mu).$$

Here,  $d_p$  is the particle diameter,  $\rho_p$  the particle density, and  $\mu$  the air viscosity (Reist, 1994). (The Cunningham slip coefficient was not used since it was not included in the STAR-CD calculations; this induces a potential error for the high-altitude case of 13% for  $5.0 \mu\text{m}$  diameter particles and 7% for  $10.0 \mu\text{m}$  particles). Since the shroud is approximately isokinetic, the shroud velocity and particle velocity can be approximated by the airstream velocity,  $V$  (and for these small angles, the horizontal component of the velocity also approximately equals  $V$ ).

The angular dependence and the exponential relationship can be taken into account by defining the length,  $L$ , as that required to straighten the particle trajectory from an initial angle of  $\varphi$  to a final angle of  $\varphi/e$ . Thus we have

$$L(\text{m}) = f[V(\text{m s}^{-1}), \tau(\text{s})],$$

and the simplest form of the relationship is

$$L = kV\tau,$$

where  $k$  is a constant equal to the ratio between the straightening distance and the quantity  $V\tau$ .

Through some exploratory plots, it was found that the best relationship (i.e., where  $k$  is actually nearly constant) was obtained by using

$$L = L_s - L_{s,m},$$

where  $L_s$  is the distance from the shroud tip to where the particle trajectory is straightened to a residual angle of  $\varphi/e$  and  $L_{s,m}$  is

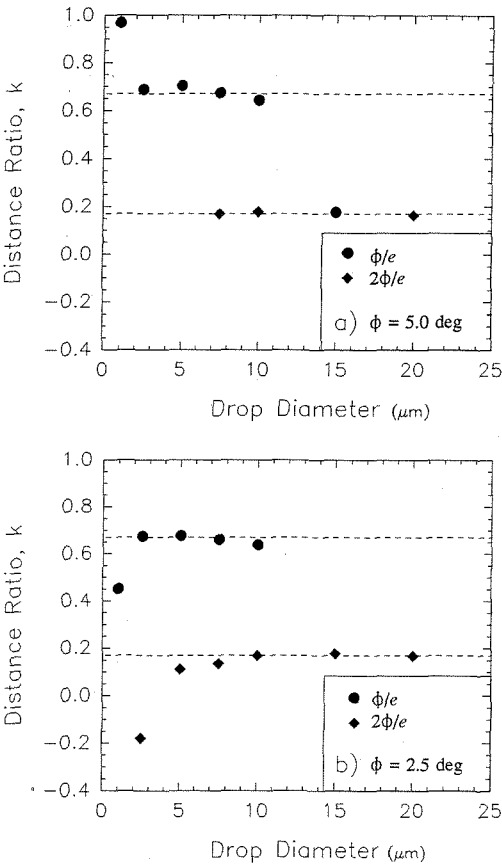


FIGURE 13. Nondimensional distance ratio,  $k$  (defined in text), as a function of particle size for a) a  $5^\circ$  incident angle,  $\varphi$ , and b) a  $2.5^\circ$  angle. Dashed lines correspond to ratios of 0.67 (to achieve a residual angle of  $\varphi/e$ ) and 0.17 (to achieve a residual angle of  $2\varphi/e$ ).

the distance from the shroud tip to where the medium (airflow) itself is straightened to an angle of  $\varphi/e$ .  $L_{s,m}$  was determined from the trajectory of a  $0.1\text{ }\mu\text{m}$  particle, assumed to follow the air streamlines.

Figure 13 shows the results of this analysis, with the distance ratio,  $k$ , plotted against particle diameter using the shroud model under DC-8 conditions. Trajectories that terminated along the centerline were used, since these can actually enter the CVI inlet. Results for  $5.0^\circ$  (top plot) and  $2.5^\circ$  (bottom plot) incident angles are shown. Since only particles less than or equal to  $10\text{ }\mu\text{m}$  in size

were straightened to a final angle of  $\varphi/e$ , results for trajectories of larger particles that were straightened to an angle of  $2\varphi/e$  are also shown. This provides at least some data for larger particles and shows that the curves approach a constant value regardless of whether  $\varphi/e$  or  $2\varphi/e$  is considered.

The distance ratio,  $k = (L_s - L_{s,m})/(V\tau)$ , is approximately constant at a value of 0.67 for a residual angle of  $\varphi/e$  and 0.17 for a residual angle of  $2\varphi/e$ . These values are the same for the  $5.0$  and  $2.5^\circ$  initial case, and thus are independent of angle. The variations at the small particle sizes are most likely due to the large uncertainty in the relatively small distances required to straighten these particles; for example, for the  $5^\circ$  case,  $1\text{ }\mu\text{m}$  particles are straightened to  $\varphi/e$  in only about five mesh cells (with three time-steps per cell), while  $10\text{ }\mu\text{m}$  particles are straightened in about 20 mesh cells. The number of steps along the particle trajectory is much smaller, and therefore  $L_s$  is less accurate, for the  $1\text{ }\mu\text{m}$  case. The determination of  $L_{s,m}$ , which is used to calculate  $k$ , is also subject to this uncertainty but is relatively small compared to  $L_s$  for the larger droplets.

The above analysis can be used as a general method to determine the shroud length ( $L_s$ ) required to substantially straighten particle trajectories, but only if the distance ( $L_{s,m}$ ) to straighten the air to a residual angle of  $\varphi/e$  can be approximated. This value was found to be relatively independent of incident angle ( $0.0105\text{ m}$  for  $2.5^\circ$ ,  $0.0096\text{ m}$  for  $5.0^\circ$ , and  $0.0092\text{ m}$  for  $10.0^\circ$ ), but will vary with the geometry of the actual shroud used. For our geometry,  $L_{s,m}$  was approximately half the shroud radius,  $R_s$  ( $0.021\text{ m}$ ), taken from the leading edge of the shroud tip to the shroud centerline. This also held true for the much smaller shroud modeled initially (Fig. 1a), where  $L_{s,m}$  was found to be about half of the shroud radius. Thus for shrouds with similar geometry to the ones used here,  $0.5R_s$  can be used to estimate  $L_{s,m}$ , and the total shroud length required to straighten particles to a residual angle of  $\varphi/e$  is approximately

$$L_s = 0.67V\tau + 0.5R_s.$$

The applicability of this relationship to lower speed applications was tested by calculating a few droplet trajectories for the low-altitude version of the final shroud model, where the freestream speed was only  $100 \text{ m s}^{-1}$ . The relationship seemed to hold reasonably well, with predicted values of  $k$  between 0.60 and 0.85 depending on droplet size and whether 100 or  $122 \text{ m s}^{-1}$  was used for  $V$ . (As mentioned earlier, the predicted speed within the shroud at low freestream Mach number was 22% higher than the freestream speed, probably due to the absence of a shock.) This result suggests that the relationship between shroud dimensions and particle straightening effects developed here should hold for a wide range of freestream conditions, as long as the flow through the shroud is approximately isokinetic and only small angles are considered.

## CONCLUSIONS AND FUTURE WORK

The following conclusions can be made from this study:

1. The STAR-CD code was useful for determining the sensitivity of the flow field to certain shroud features. In particular, the code predicts general characteristics of the flow and regions where shocks and nonaerodynamic flow patterns are likely to occur. Thus, some tuning of shroud (or inlet) geometry can be made before incurring large costs to build and test various configurations. Flow speeds in accelerating regions were overpredicted by about 4–14%, with accuracy diminishing with proximity to the shock. Some of this discrepancy is probably due to lack of mesh refinement in the boundary layer region. Wind tunnel measurements should still be used for fine-tuning if absolute accuracies greater than these are required.
2. Based on wind tunnel results, an exit to inlet area ratio of 1.34 was found to provide isokinetic flow through most of the final shroud for Mach 0.805. It was noted

that this ratio is probably larger than that required for simpler shrouds without shock regions. Flow separation did not occur for this shroud geometry at angles of attack up to  $6.7^\circ$ , and separation that occurred at  $10.0^\circ$  apparently did not affect flow along the shroud centerline.

3. The code also can be used to predict the effectiveness of the shroud in straightening particle trajectories. The relatively small shroud used in this study significantly reduced (by a factor of  $\varphi/e$ ) the incident angle of a  $10 \text{ }\mu\text{m}$  diameter, unit-density particle at 200 mb and Mach 0.805. The shroud would have to be lengthened to have a substantial effect on larger particles. A simple approximation of required shroud length as a function of particle relaxation time and velocity was developed; this should hold for a variety of conditions, assuming the shroud is isokinetic and small angles are considered. The apparent ability of the code to predict particle behavior has been demonstrated by Miloshevich and Heymsfield (1997) and Laucks and Twohy (1998).

In future work, particle collection efficiency of the shrouded CVI on the DC-8 will be compared to collection efficiency determined during earlier experiments when the CVI was flown without a shroud. This can be done by comparing the number of droplets collected by the CVI with the number above the CVI cut size measured by other probes, e.g., the Forward Scattering Spectrometer Probe (FSSP). This should provide some verification that the shroud is actually straightening trajectories as expected.

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